

Bologna Cool Routing

A dark green, stylized leaf graphic with a central vein, positioned to the right of the main title.

Heat Aware Pedestrian Routing using the TALEA Digital Twin

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The Problem



Standard Routing Today

- Only minimises distance
- Ignores heat completely
- Ignores greenery and shade
- Routes through concrete streets in full sun
- No consideration of walking comfort



What We Need

- Routes that avoid hot streets
- Routes through parks and tree-lined streets
- A small extra distance for a big comfort gain
- Especially useful in summer (35°C+)
- Important for elderly and heat-sensitive people

Bologna gets very hot in summer — but navigation apps ignore this entirely.

What is a Digital Twin?

A digital twin is a virtual replica of a real place built from real measured data.



Real Bologna

Buildings, streets, parks the actual physical city as it exists today



TALEA Digital Twin

A digital copy with real satellite measurements of heat and vegetation for every 100m square



What We Use It For

We read the heat and greenery score for each street from the digital twin and use it to route pedestrians

TALEA — The Dataset Behind Our Project

What is TALEA?

- Created by researcher SimReale at University of Bologna
- Covers the entire city of Bologna
- 11,560 grid cells each 100×100 metres
- Like a chess board of data over the city
- Free and open source on GitHub

UHEI — Heat Score

- Urban Heat Exposure Index
- Measures how hot each 100m area is
- Calculated from satellite thermal data
- Range: 0.55 (cool park) → 2.68 (hot concrete)
- City average: ~1.95
- Narrow range = compressed heat variation

NDVI — Vegetation Score

- Normalized Difference Vegetation Index
- Measures how green each 100m area is
- Calculated from satellite imagery
- Range: 0.07 (concrete) → 0.82 (dense park)
- Wide range = strong variation across city
- Stronger routing signal than UHEI

The Data Sources



TALEA — Bologna's Digital Twin

- 11,560 grid cells covering all of Bologna
- Each cell = 100×100 metres
- UHEI — Urban Heat Exposure Index
- NDVI — Vegetation Index (how green)
- Real satellite measurements
- Free and open — downloaded from GitHub



OpenStreetMap via OSMnx

- Complete Bologna walking street network
- 24,584 intersections (nodes)
- 66,156 street segments (edges)
- Real map data — not simulated
- Pedestrian walking network only
- Free and open — downloaded via Python

Our Approach

1



Get the Data

Download TALEA heat grid + Bologna street network (OSMnx)

2



Connect the Data

Spatial join: give each street its heat and vegetation score from the grid

3



Score Streets

Compute comfort cost per street: 20% distance + 40% heat + 40% (1-greenery)

4



Find Best Route

Dijkstra's algorithm finds the path that minimises total comfort cost

The Comfort Cost Formula

$$\text{comfort_cost} = 0.2 \times \text{distance} + 0.4 \times \text{heat} + 0.4 \times (1 - \text{greenery})$$

20%



Distance

We still care a little about not walking too far. But distance is the smallest factor.

40%



Heat (UHEI)

Hotter streets get a higher cost. The algorithm avoids streets in full sun.

40%



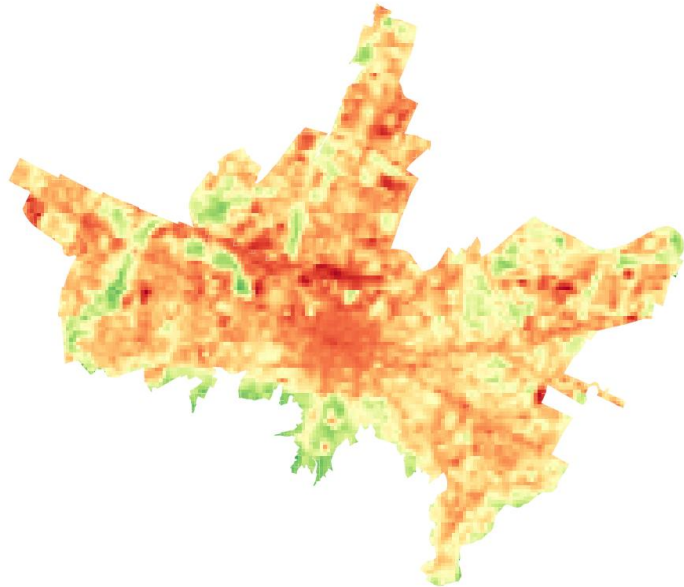
Lack of Green

(1-NDVI) flips greenery so green streets are cheap and concrete streets are expensive.

All values normalised to [0–1] before combining. Weights validated through sensitivity analysis.

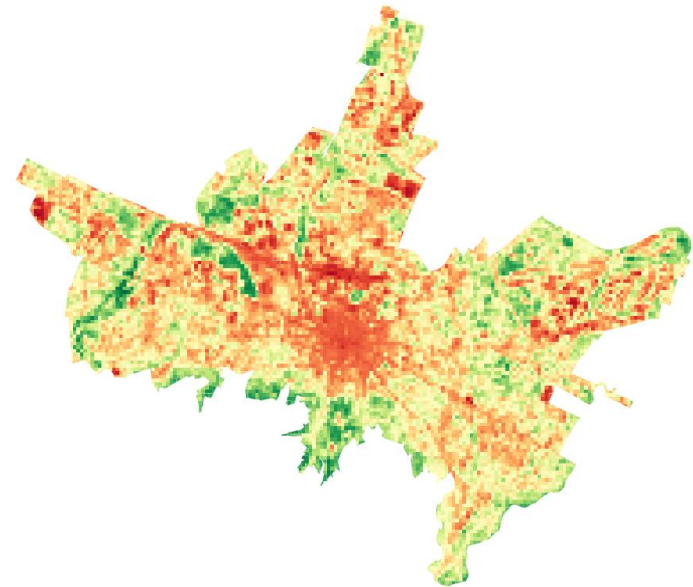
Bologna — Heat and Vegetation Maps

Heat Map (UHEI)
Red = Hot, Green = Cool



Bologna — Heat vs Vegetation

Vegetation Map (NDVI)
Green = More Plants



Left map: UHEI — Red = hot areas (north/centre)

Right map: NDVI — Green = vegetation (parks, hills)

Proof of Concept Result

Bolognina (hot, dense north) → Giardini Margherita (large green park, south)

8.7%

Greener
(NDVI increase)

2.0%

Cooler
(UHEI reduction)

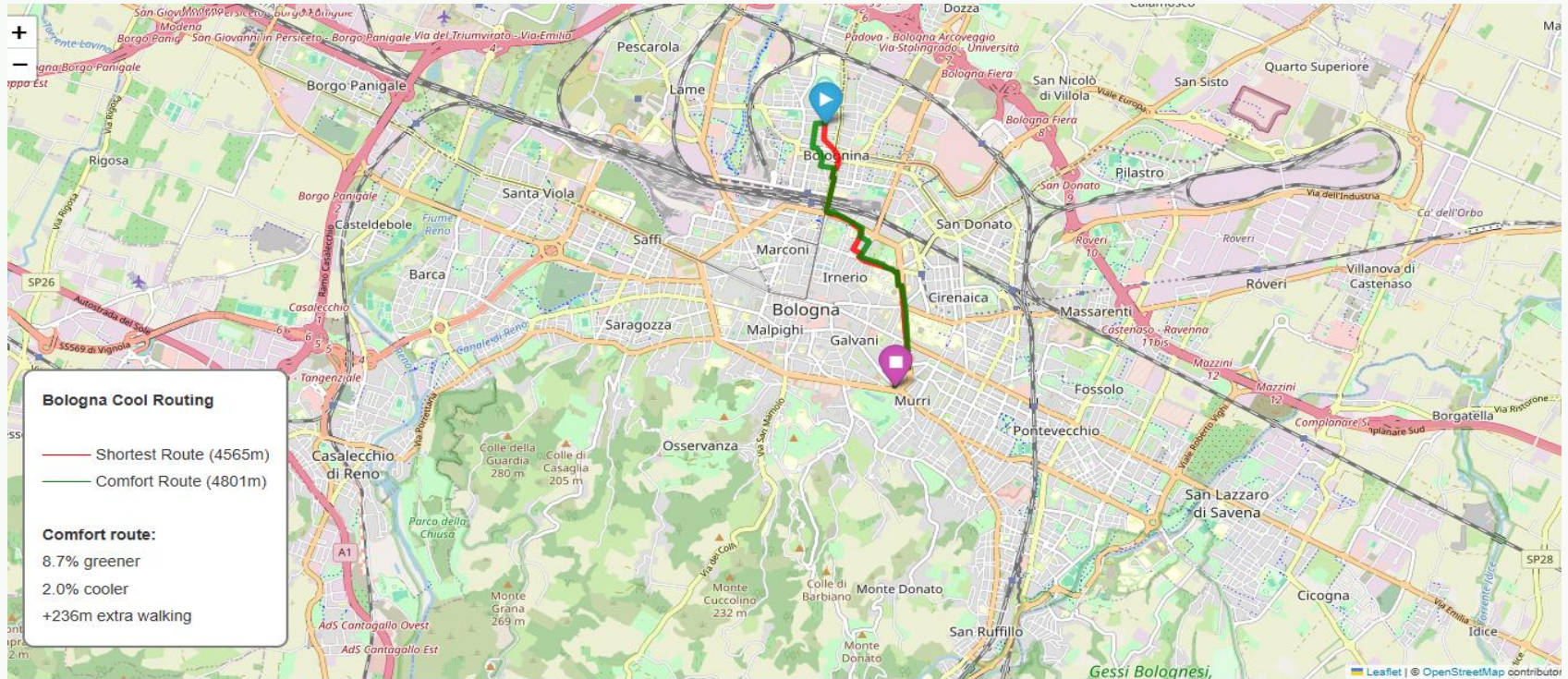
~3 min

Extra Walking
(only 236m)

Shortest: 4,565m | UHEI 2.1118 | NDVI 0.2791

Comfort: 4,801m | UHEI 2.0701 | NDVI 0.3035

Interactive Route Map



Red line = Shortest route

Green line = Comfort route

Multi-Route Analysis — 6 Corridors Tested

Route	Shortest	Comfort	Extra	Heat ↓	Green ↑
Bolognina → Giardini Margherita	4,565m	4,801m	+236m	−2.0%	+8.7%
Stazione → Università di Bologna	2,154m	2,394m	+240m	−2.3%*	−14.4%*
Piazza Maggiore → Parco Villa Ghigi	2,721m	2,981m	+260m	−1.1%	+3.4%
Fiera District → Parco della Montagnola	3,552m	4,344m	+792m	−4.7%	+20.3%
Corticella → Giardini Margherita	7,285m	9,603m	+2,318m	−5.7%	+19.3%
Borgo Panigale → Parco dei Cedri	5,442m	6,876m	+1,434m	−4.9%	+17.5%

* Route underperformed — dense city centre with no green alternatives available

Key Findings



Vegetation is the stronger signal

NDVI varies widely (0.07–0.82). Heat (UHEI) is more compressed (0.55–2.68). Green routes are easier to find than cool routes.



Longer routes perform better

Routes over 3km show 5–20% greenery gains. Short city-centre routes have few green alternatives.



One route underperformed

Stazione → Università (2.1km) got worse. Dense centre has no green streets nearby. A minimum length filter would help.



Our weights are robust

Sensitivity analysis: 3 alternative weight combinations tested. Our 20/40/40 matched or beat all alternatives.

Conclusion

Heat-aware pedestrian routing works in Bologna.

By combining Bologna's real street network with TALEA digital twin data, we built a system that finds routes that are measurably cooler and greener for only a small extra walking distance. The approach is validated across 6 route pairs with consistent results.

6

routes tested

20.3%

best greenery result

8.7%

PoC greenery gain